Semi-Autonomous Wildfire Monitoring and Control System Framework

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Industry Integrated AI-Synthesis Project

Overview and Industry Context

In recent decades, climate change has increased wildfire vulnerability in forests across many regions, including Canada (Wang 2025). As temperatures continue to rise, this threat is unlikely to decline soon. Wildfire response requires substantial resources and creates serious risks to human life (Belval 2025). This paper proposes a framework for a semi-autonomous wildfire monitoring and control system. It draws on techniques from previous projects, explaining how they are integrated and why they were selected. The paper then presents the system design, examines ethical considerations, reviews framework testing results, and discusses the project's relevance for organizations seeking AI professionals.

Integration Rationale

Three primary AI techniques were used in the design of this project:

1. **Semantic Segmentation:** We wanted a fast way to provide the system with predictions of where the fire would go next. Assuming we had either a high-altitude aerial view, or at a periodic satellite view, we built a "risk" map of the area would be a good approach. This is a simple binary classification problem for each cell in our simulation. We reused the UNet (Ronneberger 2015) built for Project 4 here.
2. **Support Vector Machine:** Aviation injuries are the primary dangers fire fighters face when fighting wildfires (Centers for Disease Control and Prevention 2014). NASA has shown that UAVs can be effective in this domain (Merlin 2009). For this simulation, we trained a Support Vector Machine to control the UAVs when they

don’t have a predetermined waypoint. SVM are fast and lightweight, making them ideal for this task. SVMs were used in a planetary science context for Project 3.

3. **Agentic AI:** A Semi-Autonomous command center was needed to provide commands to the UAVs. We chose to build this using Agentic AI. This exploits the reasoning and generative abilities of modern LLM to prioritize firefighting activities and generate waypoints for the UAVs. GraphRAG (Han 2025) was introduced to allow the system to store information about its goals and enhance reasoning abilities. A similar approach was used in Project 6.

Originally intended to use a Decision Transformer (Chen 2021) as outlined in Project 5. However, Decision Transformers require a significant number of expert training examples to make work which were not available. As such, we did not apply that technique here.

System Design and Technical Decisions

A visual overview of the system design can be found in Figure 5.

Environmental Module

This module was responsible for simulating the environment as well as generating “risk predictions” based on the environment.

Our environment was randomly generated using Perlin Noise, adapted from (Kamble 2025). We assigned environmental features: grasslands, trees, water, housing, and urban areas to numerical ranges. Figure 1 shows an example of the terrain this generates.

A similar approach was used to generate a static model for wind. To keep the simulation tractable, we assumed wind to be static once generated.

We simulated the Wildfire using a Stochastic Cellular Automata adapted from (Ghosh 2024) and (Ramadan 2024). We begin each simulation by igniting a single “cell” and letting the fire play out. Equation 1 determines the probability of fire spreading from one cell to another. An example can be seen in Figure 2.

Equation 1:

$$p_{ignite} = [1 + \exp(-k * burningtrees - windalignment - burnthreshold)]^{-1}$$

VARIABLE	**INTERPRETATION**
K	A scaling parameter.
BURNINGTREES	The number of burning cells adjacent to this cell.
WINDALIGNMENT	The dot product of the average wind direction and average fire direction about this cell.
BURNTHRESHOLD	A constant determining the minimum probability we need to “ignite” a cell.

Each burnable cell has a finite amount of fuel which decremented at the end of every step. When a cell runs out of fuel, it is converted to ash.

Given that humans sometimes head out into the forest for various reasons, we added human agents to the simulation. These human agents are stochastic and are considered "casualties" when they encounter fire.

Risk Predictor (Sub-Module of Environment Module)

We trained a modified version of a UNet to perform risk prediction. We altered the bottleneck to use multiple 3x3 convolutional layers instead of a single 1x1 convolution. Figure 3 shows a schematic diagram of our UNet.

We created a synthetic dataset using simulation output. We recorded a random starting point in the simulation, ran it for 20 steps, then recorded the output. Areas in the final image that were still burning or destroyed (but not destroyed in the original image) were marked "Risky" in the mask. An example of the results after training can be seen in Figure 4.

Model Training Parameters:

PARAMETER	VALUE
EPOCHS	10
LEARNING RATE	0.001
OPTIMIZER	AdamW
LEARNING RATE SCHEDULER	Mode=minimization, factor=0.5, patience=3

We used a hybrid loss function, as shown in Equation 2

Equation 2:

$$loss = 0.5 * BCELossWithLogitsLoss + 0.5 * DiceLoss$$

The BCEWithLogitsLoss guides the system towards handling binary classification on a per-pixel basis, while the DiceLoss helps ensure that the overall shape of the output is preserved. This helps mitigate class imbalance. Early stopping was used to avoid overfitting. Loss curves can be seen in Figure 7.

CLASS & METRIC	SCORE
SAFE – MIOU	0.9422
SAFE – DICE	0.9698
RISKY – MIOU	0.6205
RISKY - DICE	0.7564

UAV Module

We define three UAV types: Recon (a direct analogue to the UAV in Merlin 2009), Extinguishers equipped to put out fires, and SAR UAVs that can pick up humans from unsafe areas.

A hybrid approach of combining deterministic behavior with an SVM was used. When given a waypoint, UAVs proceed directly to the waypoint. If a UAVs primary action can be performed in the cell it is in, it does so automatically. If it doesn’t have a waypoint, the SVM determines its next action. This is inspired by Subsumption architecture. Connell 1990 provides an in-depth example of this approach. Figure 8 shows an overview of how these behaviors override each other.

Cells within UAVs detection radius were used as input. Detection radius varies by UAV.

We used Monte-Carlo + Q-Learning (Watkins 1992) to iteratively learn a greedy policy using a Q-Value function. A Random Forest was used to estimate Q-Value function.

An SVM was fitted to the policy generated by this.

Each UAV class has its own reward function. Reward function design was inspired by (Yu 2025) and (Julian 2018).

Recon:

$$reward = -0.5 - dist(agent_{pos}, agent_{waypoint}) + burningcellcount + 2 * humancount$$

Extinguisher:

$$reward = -0.5 - dist(agent_{pos}, agent_{waypoint}) + 25 * extinguishedfire - remaingingfire$$

Search and Rescue:

$$reward = -0.5 - dist(agent_{pos}, agent_{waypoint}) + 50 * rescuedhumancount + 5 * detectedhumancount$$

Constants were chosen via intuition with minor tuning during end-to-end testing. Further optimization here requires more end-to-end testing which is prohibitively expensive.

We performed 10 training rounds, 100 episodes per agent. Results are below:

UAV	MEAN REWARD	STANDARD DEVIATION
RECON	-5.99	6.05
EXTINGUISH	-6.9	11.05
RESCUE	-6.02	4.36

UAVs have functionality to facilitate interaction with the Command Module. UAVs keep a record of all observations / positions / waypoints for each step. Data is sent to the Command Module after an appropriate number of steps have passed. UAVs accept commands from the Command Module.

To keep the UAVs within the grid, we implemented a special “boundary override action” that automatically overrides the current input steering it in the opposite direction.

Command Module

We used Agentic Workflow as the basis for our Command Module. Figure 6 provides a schematic diagram of the workflow.

Each Agent other than the Command Centre Agent keeps a memory of previous interactions which are inserted as part the prompt for each turn. Memory is implemented as a deque. To avoid exceeding the input token limit on our model, we truncate the input. Historical data is fed to the agent in reverse chronological order.

Long term goals (LTG) are tracked using a Knowledge Graph, adapted from (Yang 2025). We have three LTG’s: Search and Rescue, Housing Protection, and Fire Fighting. Our KG ontology is in Figure 6.

MGraphDB[1] was used for GraphDB. We created tools for the Agents to add/remove/query entities and relationships from the graph.

AGENT	PURPOSE	KNOWLEDGE GRAPH ACCESS
Command Centre Agent	Orchestration. The command center agent takes in the inputs from the simulation and outputs the final response from the Workflow.	None
Strategy Agent	Generates Subgoals for LTGs	Read/Write
Dispatch Agent	Translate the activities and priority list generated by the Strategy Agent UAV Commands	Read
Assistant Agent	Responds to user queries and updates the LTG in the Knowledge Graph as needed.	Read/Write

Integration worked as follows:

[1] mgraph-db · PyPI

1. On the first step and every 40-simulation step afterwards[2], the simulation calls the Command Center Agent. The simulation provides the recent history of each UAV, a screenshot of the current area, and output from the UNet model.
2. On the first invocation of the Command Center Agent, and every 5 iterations afterwards, the Strategy Agent is invoked to update LTGs.
3. The most recent priorities and inputs are sent to the Dispatcher Agent. The dispatch agent produces a new waypoint for each UAV under its control. A reason for each command provided.
4. These commands are passed on to the UAVs.

The Operator can interrupt the simulation while it is running to query the current state. These queries are routed to the Assistant Agent, which answers questions and makes updates to the Knowledge Graph as needed.

The simulation ends either after 750 steps, or when all fire has been extinguished.

[2] Optimizes runtime and token usage.

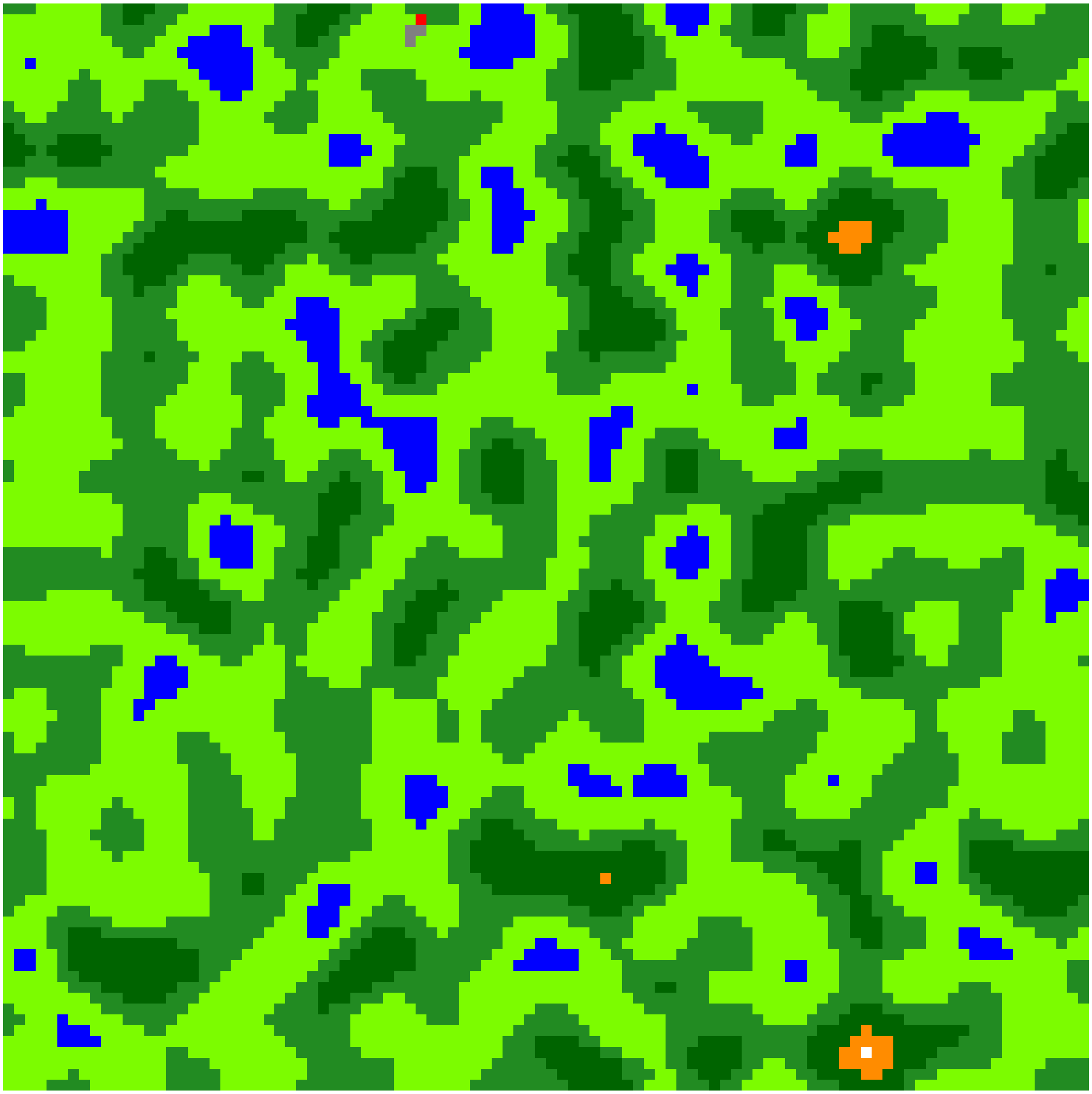

Figure 1 Perlin Noise Generated Terrain

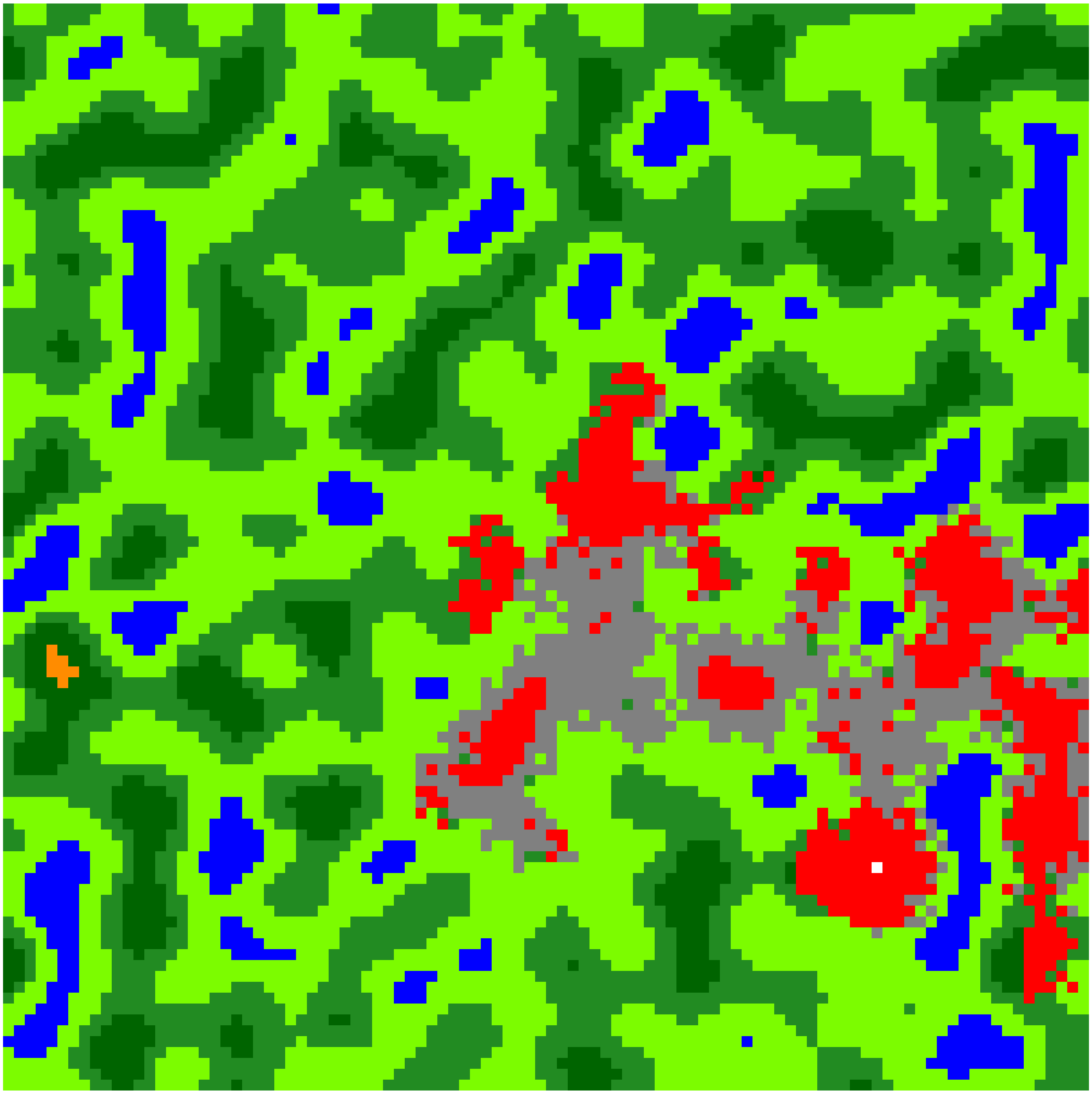

Figure 2 Wildfire CA in action

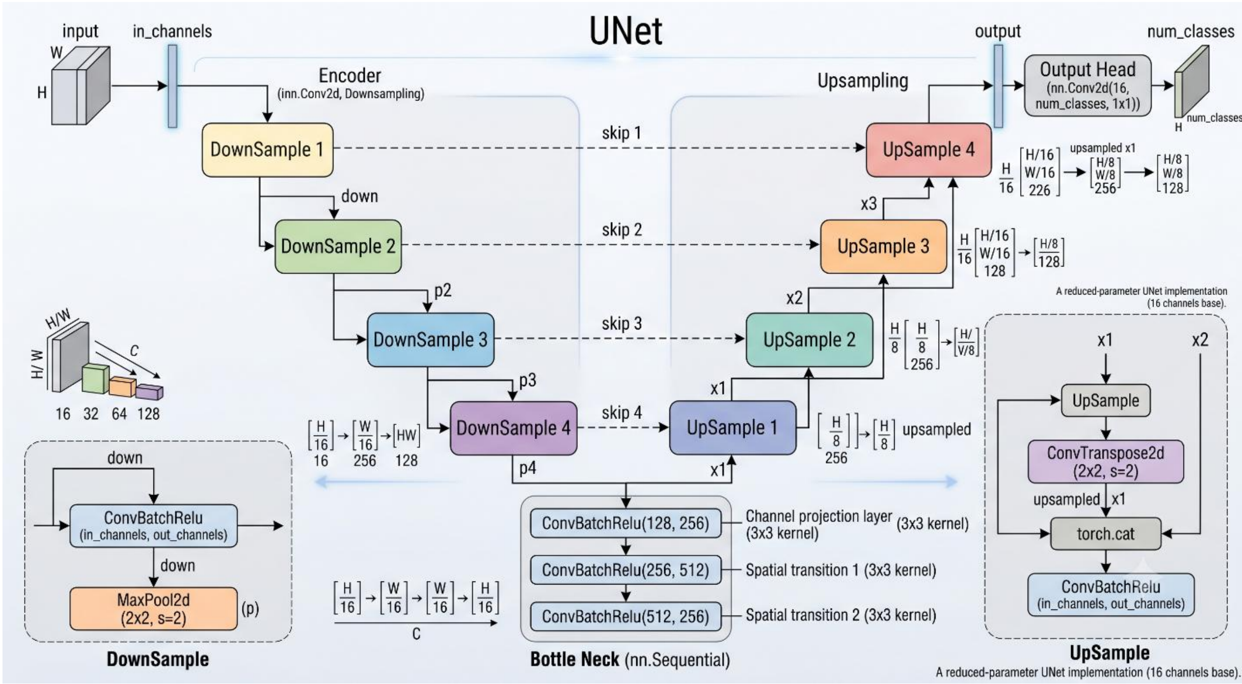

Figure 3 Modified UNet (image generated with assistance of Gemini AI Model based on authors UNet code)

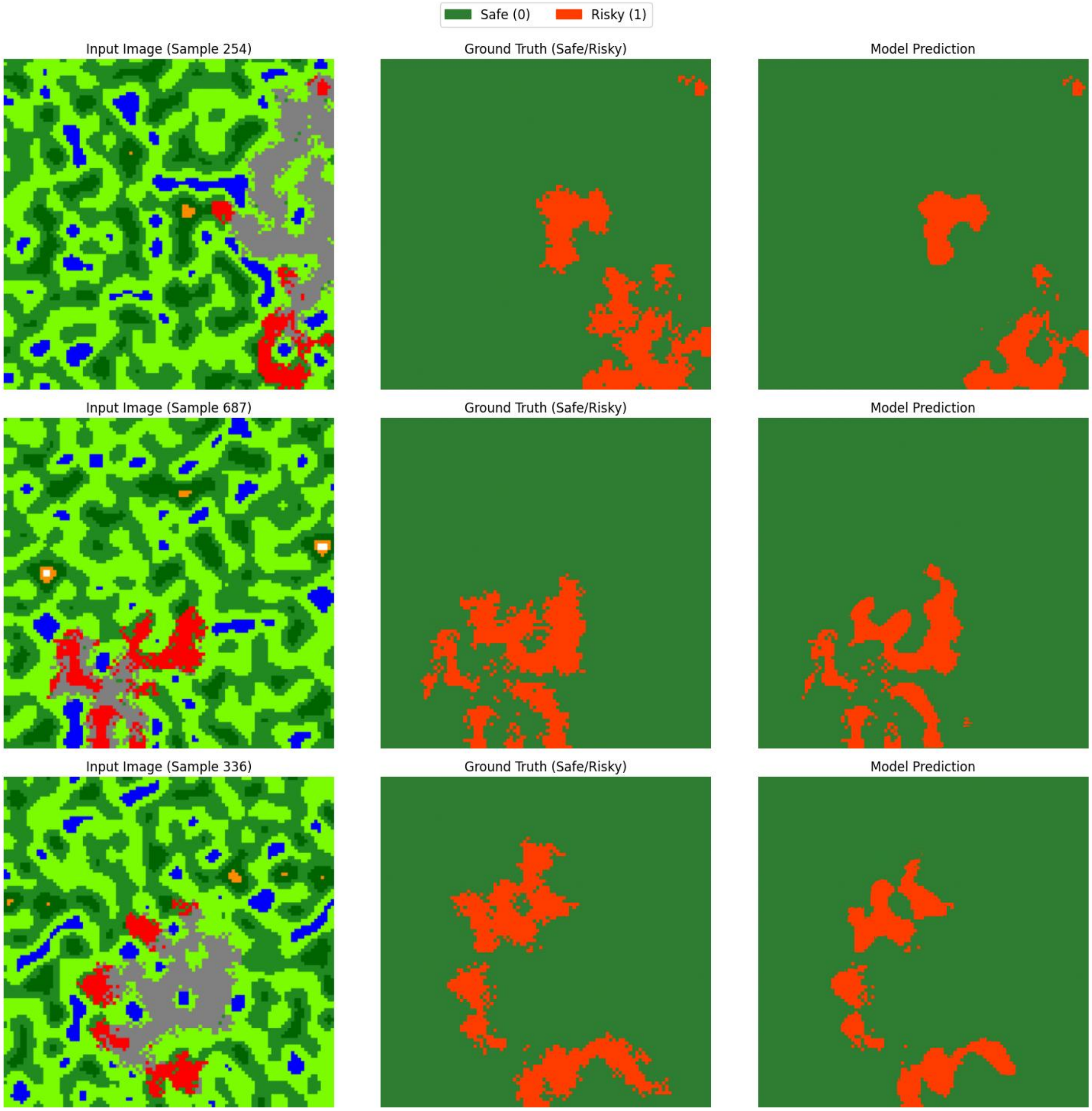

Figure 4 Fire Risk Prediction Outputs

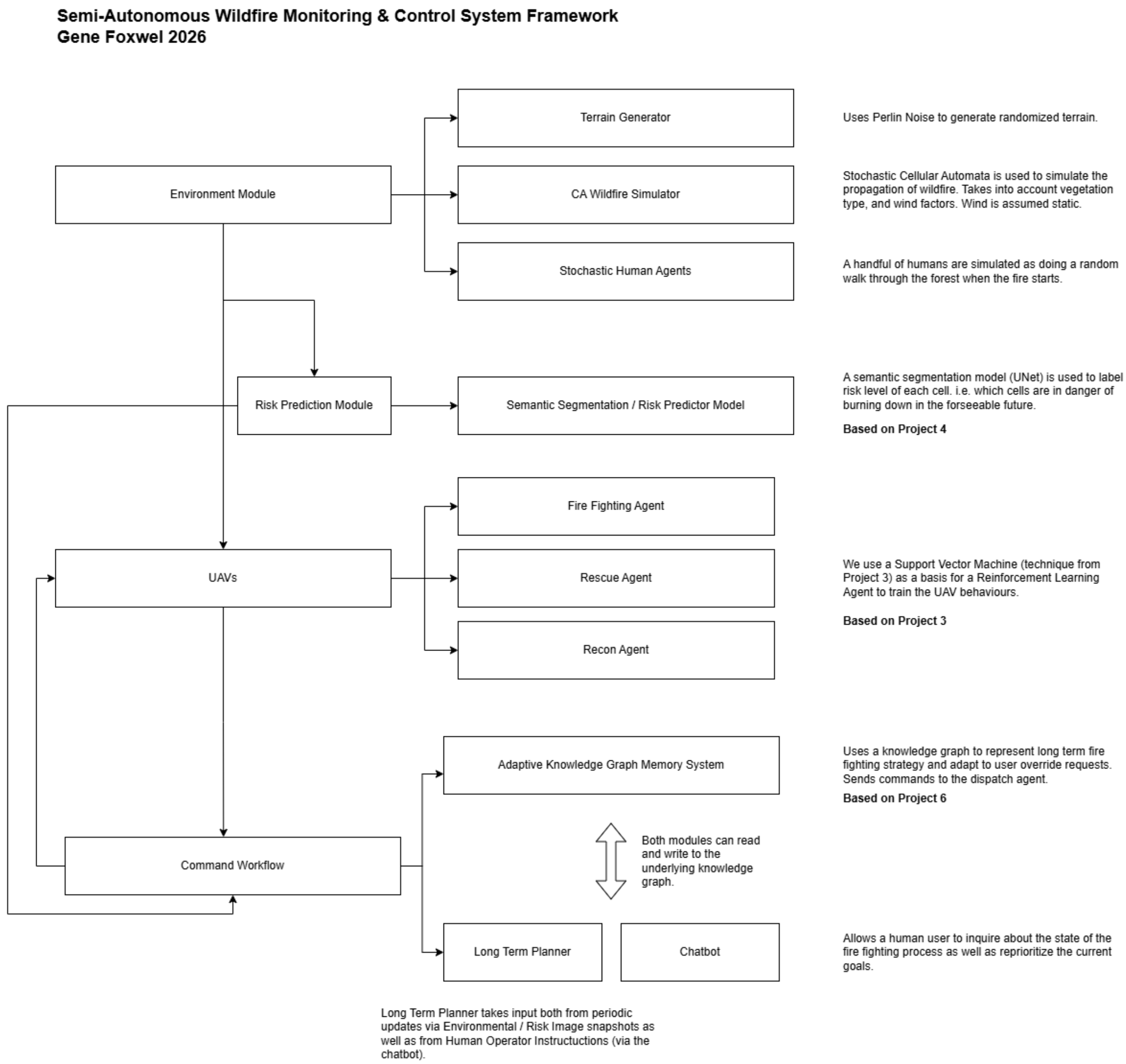

Figure 5 System Overview

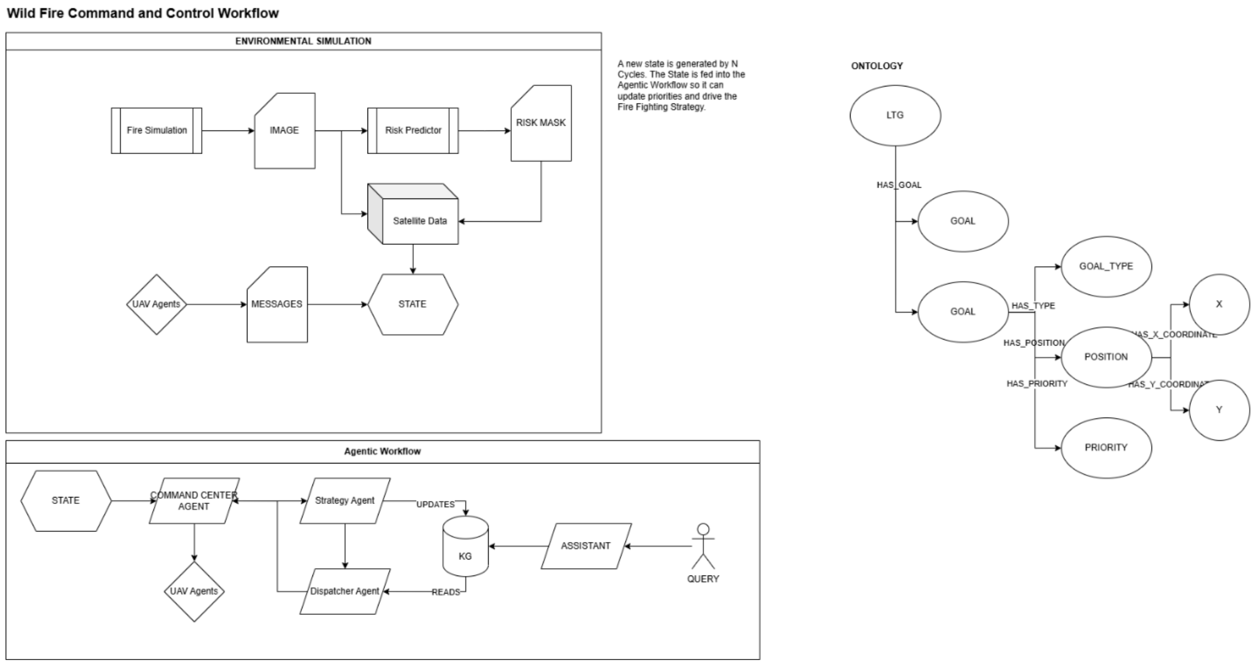

Figure 6 Agentic Workflow

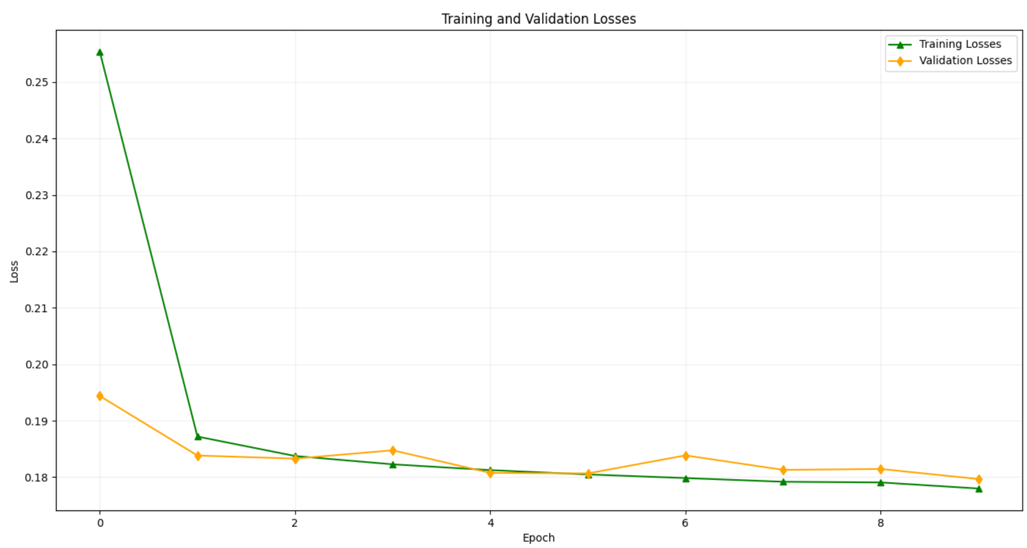

Figure 7 UNet Training and Validation Loss

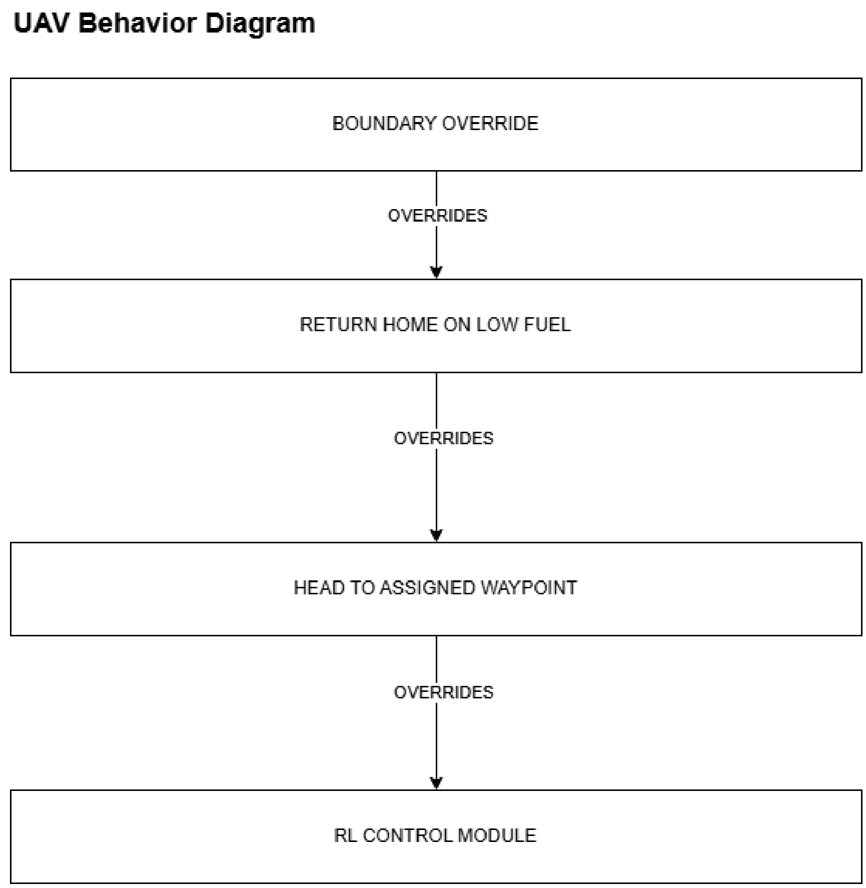

Figure 8 UAV Behavior

Ethical Considerations and Responsible AI

While fighting wildfires is a dangerous occupation, it still provides employment. Replacing these workers could leave them without a comparable source of income. This is not an intended outcome of this framework. This work and similar work should augment human abilities.

We do not recommend extending this system to be fully autonomous. Agents can sometimes fixate on unexpected things. An Agent that continues to search for a dead human while ignoring others (as happened during testing) is not a desirable situation.

Assumptions were made during testing that may not be realistic.

Examples:

1. Static wind field. Wind is rarely static. We assumed reliable aerial views of the fire.
2. Firefighting and SAR drones could be built reliably. Furthermore, especially for SAR, a human presence is preferable to deal with first-aid or psychological distress that might occur during an emergency.

Who is responsible for loss of life / property damage that may occur due to a system like this? A malfunction or a choice could lead to loss of life or property. We left in the human-in-the-loop option to override the system, but people will still defer to the computer, especially when stressed. Protocols will need to be in place to review the reasoning presented by the system to ensure they meet the expected standards.

Our setup used a remote model – Anthropic Claude Sonnet 4.5. This was sufficient for our purposes. Infrastructure may not be reliable in a realistic scenario. A local model should be used (for example Llama 4 Scout, Minstral Small[3]) so the system is independent of local infrastructure.

Evaluation and Reflection

We ran five end to end tests on the current version of the system. Despite the stochastic nature of the system, we kept the number of tests low as running the system with an appropriate model was cost prohibitive.

Each test ran for maximum of 750 steps. We recorded video and output logs for each simulation.

We used Claude Sonnet 4.5 as our LLM for all Agents.

[3] The Best Open Source and Open-Weight LLM Models to Run Locally in 2026

	STEPS	ALIVE HUMANS	RESCUED HUMANS	BURNING CELLS
EX 1	750	7	3	375
EX 2	72	13	2	0
EX 3	750	2	0	1579
EX 4	750	1	2	1590
EX 5	750	7	0	1585

Reviewing the video for experiment 2 noticed that the system obtained better performance than the others. Our video shows we had a lucky start in that the fire ignited near enough to home base allowing the fire to be put out unusually quickly.

This data collected is not representative of a full picture of what this system is capable of. These runs primarily give us qualitative information on the systems' failure modes rather than a full picture of the system.

Observations based on videos:

Firefighting UAVs fight the fire in their area but will ignore any fire nearby
Recon UAVs appear to be deployed in clumps near the fire's frontier
SAR operations seem to fixate on saving a particular human.

We had an LLM find any patterns in the output logs. We checked these to ensure accuracy.

LLM Observations:

RESCUE UAV target stickiness: In ex3 and ex5 (both 0 rescues), the RESCUE UAV was locked onto the same coordinates for 8–10 consecutive dispatch cycles

Out-of-bounds coordinates: In ex1, pixel (220, 760) got converted to grid (44, 152) — Y=152 is way off the 100×100 grid.

Speculative RESCUE deployment: The RESCUE UAV is frequently sent "just in case" to housing areas with language like "once RECON identifies civilians"

Knowledge graph goals never expire: stale goals persist indefinitely in the graph.

LLM results suggest we could improve goal tracking:

1. Allow Goal nodes to expire if not deleted after a certain number of iterations.
2. Prioritize newer goals over older goals.

We leave these improvements for future iteration.

Professional Relevance

This project demonstrates several techniques in Artificial Intelligence:

1. Semantic Segmentation as a predictive technique – which could have potential applications in Autonomous Driving (Feng 2021) and Robotics (Hurtado 2022) (as well as the original application of UNet to medical imagery).

2. Reinforcement Learning (RL) as well as Machine Learning via SVM. RL has already been applied to UAV control in this domain (Julian 2018). RL has applications to robotics (Rajeswaran 2017) and Automated Gameplaying (Mnih 2015).
3. Agentic Workflows, which is finding applications in Finance (Yang 2024), and Entertainment (Ammanabrolu 2020).

We combined these to model Wildfire Monitoring Control, demonstrating the ability to synthesize multiple techniques into a single system.

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